

AudioEval: Music Score-based Instrument Performance Assessment System

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Abstract

Useful practice feedback identifies an error, locates it in the score, and lets the student replay the relevant passage. We present AudioEval, a layered approach combining transcription, score alignment, and error labelling. Transcription estimates the notes in a recording; score alignment associates them with the intended written notes. The current AudioEval uses a dual-resolution CTC transcriber, retrieves the practised passage, and aligns candidate notes using bounded repetition and ornament templates. Acoustic verification supports note-error decisions, and uncertain alignments are explicitly withheld. Synthetic performances supply training examples with known relationships between played and written notes. In the previously evaluated Basic Pitch-based configuration, AudioEval achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, compared with 41.39% for Polytone and 34.71% for LadderSym. This measure assesses both score correspondence and event type across correct, repeated, and erroneous notes. We retain the earlier comparison on 11 manually labelled clarinet recordings and two case studies; these results do not evaluate the revised pipeline. We present the design of MusicPal, a proposed desktop practice companion for recording performances, displaying AudioEval’s feedback, and replaying the relevant passages.

1 Introduction

A student practising a musical instrument alone needs to know which notes require attention and where to find them in the score. A useful feedback system should identify the type of error, highlight its location, and let the student replay the corresponding passage.

Recent error-detection methods learn to compare a performance with its intended reference. Polytone and LadderSym use synthetic musical audio and scores to train this comparison at a scale that would be difficult to obtain through manual annotation [4, 5]. LadderSym also reports evaluation on real beginner piano performances.

Their output categories are correct, missed, and extra notes [4, 5]. A return to an earlier passage can therefore appear as a sequence of extra notes without identifying which passage was repeated. When a student returns to an earlier passage, the same written notes occur again at different times in the recording. Interpreting such performances requires acoustic evidence to be considered in its musical context: identifying what was played is closely connected to determining where it belongs in the score.

AudioEval combines transcription, score alignment, and error labelling to make this connection. Its current acoustic network learns clarinet note sequences from dual-resolution spectrograms while retaining uncertain candidates. A passage locator connects partial recordings to a full score, and a bounded dynamic program compares repetition- and ornament-aware templates using pitch, confidence, and onset timing. Separate acoustic checks verify possible extra and missed notes. The resulting feedback identifies affected score notes and replay intervals, with an explicit uncertain state when alignment is unreliable.

The earlier Basic Pitch-based AudioEval configuration achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, compared with 41.39% for Polytune and 34.71% for LadderSym. The comparison uses common recordings, reference annotations, and a scoring rule that rewards both correct location and correct event type. Section 5 describes the measure and the training conditions of the compared systems. These historical results are preserved for provenance; a matched comparison of the revised method in Section 4 remains to be run.

We also designed MusicPal, a desktop practice companion for recording performances and reviewing AudioEval’s feedback. MusicPal would record a performance, request analysis from AudioEval, and present the returned feedback on the score and through audio playback. Saved recordings could later be manually annotated and used for model development.

The contributions are:

1. AudioEval, a layered method combining clarinet transcription, bounded passage alignment, explicit repetition tracking, and verified score-linked feedback, together with a documented comparison of its earlier configuration against two existing detectors.
2. The design of MusicPal, a practice companion for recording, displaying feedback, and retaining practice data for future manual annotation.

2 Related Work

2.1 From Transcription to Practice Feedback

Automatic music transcription estimates which notes are present in a recording and when they occur. Models such as Onsets and Frames, MT3, and Basic Pitch provide different approaches to this task [2, 7, 9]. Practice feedback requires an additional reference: the score the player intended to perform. The same detected pitch can be correct in one passage and a wrong note in another. Locating the sound in the score is therefore part of interpreting the performance.

Score-informed detectors combine acoustic evidence, alignment, and note comparison [1, 15]. Dynamic time warping aligns two sequences while accommodating changes in speed [6]. More specialised approaches address local performance errors, repeats, and skips [12, 13]. RUMAA brings transcription, score alignment, and mistake detection into a unified framework with repeat handling [3]. The current AudioEval combines a clarinet-specific CTC transcriber with passage retrieval and bounded repeat templates, retaining alternative note candidates during alignment.

2.2 Learned Error Detection

Polytune learns to compare practice audio with audio rendered from the reference score and predicts correct, missed, and extra notes [4]. LadderSym extends this approach with repeated interaction between the two input streams and a symbolic score prompt [5]. Both systems learn to predict error labels from the practice and reference inputs.

AudioEval represents the correspondence between played and written notes explicitly. Repeat templates preserve canonical score identity and performed occurrence; the current feedback exporter reports passage repetition separately from first-pass content errors. The existing comparison with Polytune and LadderSym evaluates the earlier AudioEval configuration as a complete system. Because their original outputs describe timestamped notes, a common conversion assigns their predictions to score locations for the evaluation used here.

2.3 Synthetic and Real Performance Data

Large transcription datasets provide detailed note annotations: MAESTRO pairs piano audio with MIDI, while MusicNet and URMP cover additional instrumental settings [10, 11, 14]. Such annotations describe the notes that were played. Error detection also needs the intended score and an account of deviations from it.

Synthetic generation supplies these relationships by modifying a score before rendering its performance as audio. Polytune and LadderSym use this approach for error supervision, including instruments represented in CocoChorales [4, 5, 16]. Their synthetic datasets support controlled comparisons with known errors. LadderSym additionally evaluates on a collection of 20 real beginner piano recordings [5]. Our synthetic generation preserves the score location of each altered and repeated note to supervise score-linked clarinet practice feedback. The proposed MusicPal device would allow future practice recordings to be saved for manual annotation.

3 Synthetic Training Data and Design Rationale

3.1 Synthetic Training Performances

We generate labelled training performances by introducing wrong notes, missed notes, extra notes, rhythm changes, and unintended repetitions into a reference score. The modified performance is then rendered with a clarinet SoundFont, a sampled-instrument sound library. During generation, we record how each performed event relates to the original score.

For example, a wrong note changes the played pitch while preserving the intended written location. A missed note remains a target in the score although it produces no sound. An extra note is recorded at its position in the generated performance but has no written counterpart. When a passage is repeated, each performed occurrence retains its link to the original written notes. These relationships provide training targets for both score alignment and error labelling.

The historical Basic Pitch-based AudioEval training set contains 4,544 synthetic recordings. The validation set contains 358 recordings and is separated from training by source work and score content, with duplicate score excerpts grouped before partitioning. Pitch conventions, note timing, and correspondences between generated notes and the reference score are checked before use. During inference, the detector receives only the recording and reference score; the generation records are used to construct training targets and evaluation annotations. The current acoustic checkpoint additionally uses the realistic, fast-note, and practice-like synthetic families recorded in Appendix A.1; the historical counts above are not its training-corpus totals.

3.2 Design Rationale

An error label is useful during practice when the student can locate the note in the score and replay the corresponding sound. AudioEval therefore keeps track of score locations and recording intervals as it identifies note errors and unintended repetitions.

We train a clarinet-specific transcriber with complementary time resolutions to represent both pitch structure and short attacks. AudioEval retains uncertain note candidates for score-informed selection, retrieves the attempted passage, and compares bounded restart hypotheses. Separate error-verification models help distinguish a performance mistake from a transcription failure. Table 1 summarises these choices and the alternatives considered.

Table 1: Design choices in AudioEval and MusicPal, their alternatives, and their intended purposes.

Design aspect	Alternatives	Selected approach and purpose
Acoustic recognition	Train a task-specific transcriber; reuse a pretrained transcriber	Train a dual-resolution CTC network with auxiliary pitch and boundary supervision.
Note selection	Commit to one transcription; consider alternatives during alignment	Retain primary, optional, and alternative-pitch candidates for structured score alignment.
Score alignment	Require forward progress; allow returns to earlier passages	Retrieve a bounded passage and compare repeat and ornament templates with timing refinement.
Error decisions	Predict all labels directly; interpret a shared correspondence	Verify extra and missed proposals acoustically; report canonical note identities and uncertainty.
Device processing	Run analysis on the device; use a remote server	Use the device for recording and presentation and the server for AudioEval.

4 AudioEval

AudioEval takes a monophonic clarinet recording and a MusicXML score and returns score-linked wrong-note, missed-note, extra-note, and unintended-repetition labels. The current implementation combines a trainable acoustic transcriber with a structured score aligner: acoustic evidence proposes what was played, and the aligner determines which score events explain it while allowing genuine departures from the notation. Figure 1 summarises the inference path.

This section describes the current dual-resolution transcriber and bounded passage aligner. The earlier Basic Pitch-based system is retained in Appendix A.2 to document the existing experiments. Those results have not been rerun with the current stack. The default application uses the passage-enabled v9 configuration; the v10 boundary refinement described below is a separate experimental extension. Neither configuration’s deployment status establishes an accuracy improvement over the historical system.

4.1 Score Representation and Acoustic Input

We parse the intended part into a canonical sequence $S = (s_1, \dots, s_M)$, where each event stores its written MIDI pitch, onset and duration in quarter-note units, measure, part, and unique score index. Tied notes form one sustained event; separately articulated notes retain separate identities even when their pitches agree. These identities distinguish two occurrences of the same melody in different score locations. Ornament annotations are retained for the aligner’s local performance templates.

Audio is converted to mono at 22,050 Hz. The current acoustic model predicts written pitches for B-flat clarinet: sounding pitch is two semitones below written pitch. Training loaders use the recorded pitch-space metadata; the deployed configuration assumes a written B-flat-clarinet score. Concert-pitch inputs therefore require an explicit conversion. Inference uses the reference MusicXML and recording, without access to the modified synthetic score or the generation-time note correspondence.

4.2 Dual-Resolution Acoustic Transcription

The transcriber is a clarinet-specific neural network trained on synthetic performances. A long analysis window preserves pitch structure, while a short window retains attacks and brief notes.

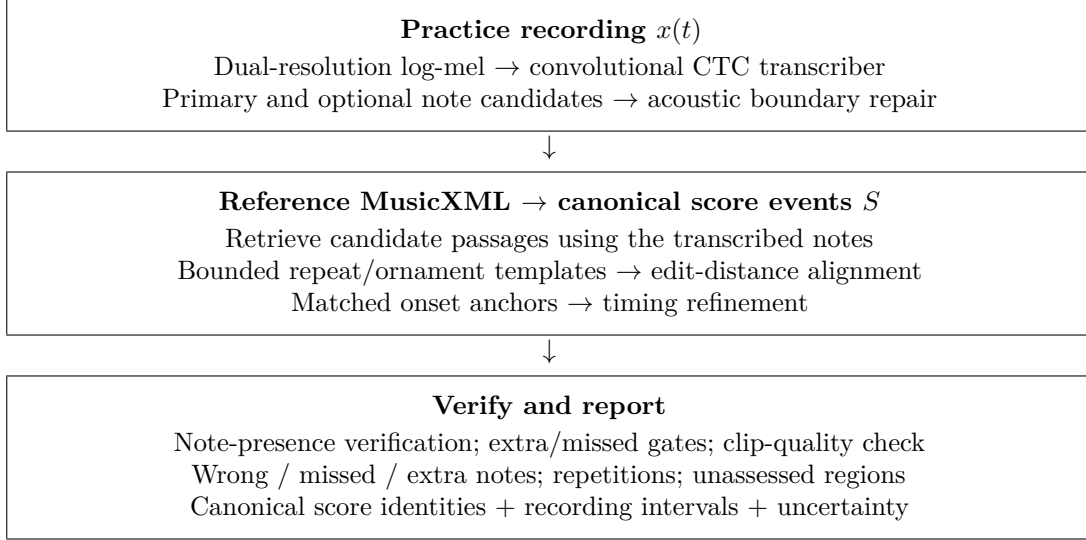


Figure 1: Current AudioEval inference. The transcriber uses only the recording. Score context enters through passage retrieval, structured alignment, and verification of possible errors. The optional v10 refinement restores already decoded same-pitch boundaries and reruns alignment under a consistency check. Error identities refer to the original score; recording intervals support replay.

Both log-mel spectrograms use a 256-sample hop, approximately 11.61 ms. The long branch has a 2,048-sample window and 128 mel bands; the short branch has a 512-sample window, a 1,024-point Fourier transform, and 64 mel bands. Each branch is normalised using recording-level quantiles. Their frames share the same time origin.

Separate convolutional encoders compress the frequency axis while preserving time resolution. Their outputs are concatenated, projected to 256 channels, and added to a projection of the original 192 mel bands. Twelve residual temporal convolution blocks use kernel width five and dilations 1, 2, 4, 8, 16, 32, repeated twice. This supplies approximately 2.9 seconds of convolutional context on each side of a frame. The network outputs a connectionist temporal classification (CTC) distribution over written MIDI pitches 52–100 and a blank symbol, together with auxiliary estimates of pitch, voicing, onset, boundary, rearticulation, and confidence. Long recordings are processed in overlapping windows whose probabilities are averaged with tapered weights.

CTC [8] represents an ordered note sequence without requiring each output note to occupy its full acoustic duration. For a target pitch sequence Y and acoustic features X , the sequence objective is

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi: \mathcal{B}(\pi)=Y} \prod_{t=1}^T P_{\theta}(\pi_t | X), \quad (1)$$

where $\mathcal{B}$ collapses consecutive identical tokens and removes blanks. Distinct successive notes of the same pitch require a separating blank in the CTC path. The auxiliary boundary targets provide additional supervision for these rearticulations.

Decoding scales the blank probability before taking framewise maxima and collapsing runs. Each primary candidate retains its pitch, onset, peak CTC support, and an alternative pitch with its support. Additional non-blank probability peaks in otherwise uncovered regions form optional candidates. Brief intermediate pitches with weak onset and rearticulation evidence can also be marked optional. The aligner may retain an optional candidate if the score supports it, or discard it; short duration alone does not remove a note. Candidate end times are initially estimated from the next onset, so they are playback approximations rather than independently measured offsets.

4.3 Adjacent Same-Pitch Boundaries

A sustained note can produce two CTC emissions, while a genuine repeated note can have little silence between attacks. The v9 postprocessor examines adjacent equal-pitch candidates using local energy and spectral continuity. A separate waveform envelope uses centred 4 ms RMS windows at 1 ms intervals, allowing the gap check to inspect interruptions shorter than the mel hop. A detected low-energy core with stronger flanks prevents merging; missing or insufficient evidence also retains the boundary. The permissive v9 continuity settings can nevertheless merge real rearticulations. Absence of a detected gap is not evidence that the performer played only one note.

The experimental v10 extension addresses a restricted subset of these cases. After an initial alignment, it considers a merged group only when its original, confident CTC candidates correspond to an equal number of consecutive, distinct, same-pitch score events. A small random forest combines gap, energy, spectral, attack, and voicing features to vote on whether to restore an original boundary. It does not invent an onset. The proposed transcription is realigned and accepted only if the restored candidates map to separate consecutive score events and mappings, deletions, and error identities outside the affected groups remain unchanged. Otherwise the original result is returned. This check limits changes outside the proposed repair; it does not prove that a restored boundary is acoustically correct.

4.4 Locating a Practised Passage

A recording may cover only part of a long score. We first retrieve plausible passages using confidence-weighted subsequence edit distance between reliable transcribed pitches and the canonical score. Unmatched score prefixes and suffixes have no cost, so notes outside the attempted passage are not automatically treated as omissions. Two rolling dynamic-programming rows keep retrieval memory linear in score length.

The locator retains up to four distinct passages and adds eight context notes at each end. Each window is then evaluated by the detailed aligner below. Its ranking combines alignment cost per transcribed candidate with retrieval dissimilarity. When several locations remain close to the best candidate and pass the alignment-quality threshold, the current runtime selects the earliest plausible location and records the ambiguity. This is a deterministic convention for equivalent passages, not proof of the performer’s intended location. Short supplied excerpts bypass retrieval. Insufficient evidence or a search exceeding the configured limits produces an uncertain result.

4.5 Repetition- and Ornament-Aware Alignment

For each retrieved passage, the aligner constructs a bounded set of performance templates. Linked units retain their original score index and an occurrence index; additional ornament units carry their expected pitch and occurrence without inventing a canonical score note. The no-restart template competes with templates containing one or two additional copies of a passage. Local proposals come from consecutive repeated phrases in the transcription and can represent two non-overlapping restart regions. Measure-aligned repeat proposals fill the remaining search budget, prioritised by agreement with the observed sequence length. Written ornament patterns allow supported embellishment notes and linked score units to be considered together. The search does not cover arbitrary nested repeats or unrestricted jumps.

For example, the written sequence 1, 2, 3, 4 and performance 1, 2, 3, 2, 3, 4 can be explained by a template that copies events 2–3 and then resumes at event 4. Both visits preserve the same canonical score identities, while occurrence indices distinguish their recording intervals. A repeated motif already present in the written sequence can instead be explained by the no-restart template.

For a template U_h , dynamic programming minimises edit costs over acoustic candidates C . Permitted operations include pitch match or substitution, insertion of an extra candidate, deletion of a score unit, rejection of an unreliable candidate, selection of a supported alternative pitch, and permitted ornament or same-pitch span transitions. A schematic recurrence is

$$D_h(i, j, q) = \min_{o \in \mathcal{O}(i, j, q)} [D_h(i - \Delta_i(o), j - \Delta_j(o), q_o) + c_o(C, U_h)], \quad (2)$$

where q is the operation state and c_o is the configured cost. The start state has cost $D_h(0, 0, 0) = 0$; unreachable states have infinite cost. The state preserves constraints on linked units and merged spans. Candidate confidence affects insertion and rejection costs, and an additional penalty discourages unnecessary repeat copies. These are explicit edit costs, not a learned neural path score.

The first pass selects the lowest-cost repeat template. Confident pitch matches then anchor a map from the template’s quarter-note positions to recording time. If at least four anchors are available, a second dynamic-programming pass on the selected template adds a capped onset-deviation cost,

$$c_{\text{time}}(i, j) = \lambda_t \min \left\{ \left(\frac{t_i - \hat{t}_j}{\tau_j} \right)^2, \kappa \right\}, \quad (3)$$

where $\hat{t}_j$ is the predicted onset and τ_j depends on expected note duration with a fixed lower bound. This pass helps resolve repeated-pitch ambiguities while allowing expressive timing. It refines the selected template rather than optimising timing jointly over every repeat hypothesis. Window-local indices are finally mapped back to the original full-score indices.

4.6 Error Verification and Score-Linked Feedback

The alignment proposes substitutions, insertions, deletions, and repeated passages. A pitch substitution on the first pass gives a wrong-note label. An unexplained retained candidate gives a possible extra note, and an unfilled first-pass score unit gives a possible missed note. Repeat-copy indices identify the repeated passage and its additional occurrences; the current feedback exporter does not issue separate content-error labels within replayed copies.

Extra and missed proposals pass through separate logistic classifiers using acoustic support, duration, neighbouring alignment, local timing, and structural context. Missed-note verification also queries a small convolutional network on the dual-resolution spectrogram around the expected onset, conditioned on harmonic templates for the expected pitch and its neighbours. It tests whether a note may have been played but omitted by the transcriber. Low-support error calls are withheld. Gate scores are operating-point scores and are not assumed to be calibrated error probabilities.

A clip-level check withholds all error labels when the fraction of alignment events marked as matches or repeat copies is too low. This heuristic includes replay events and is not a calibrated accuracy estimate. Retrieval failure and resource-limit failures also return an explicit uncertain status. Unplayed score regions and withheld decisions remain unassessed; an empty error list must not be interpreted as a verified clean performance.

Each emitted label carries canonical affected-note identities, separate contextual score notes, and a recording interval. Extra notes are anchored to neighbouring score events. Missed-note playback times are interpolated from surrounding aligned onsets, with a tempo-map estimate as fallback and an explicit unavailable state if timing cannot be estimated. Repetition labels include the replay interval, the first-pass interval, and the number of additional copies. The current stack uses timing to improve alignment but does not emit rhythm, intonation, or timbre-error labels. The historical rhythm classifier belongs to the earlier experimental system.

4.7 Training and Configuration Selection

Synthetic generation supplies both the performed note sequence for acoustic training and its relationship to the clean score for alignment and error verification. Transcription targets describe the notes actually rendered, including wrong pitches, inserted notes, and repeat copies. This prevents the acoustic network from being trained to reproduce the intended score when the performance differs.

The current transcriber combines the CTC loss in Equation 1 with a weighted frame-level objective,

$$\mathcal{L}_{\text{acoustic}} = \mathcal{L}_{\text{CTC}} + 0.5\mathcal{L}_{\text{frame}}. \quad (4)$$

The frame objective supervises pitch, voicing, onsets, boundaries, rearticulations, and confidence, with additional within-note consistency terms. Feature-space augmentation varies gain, spectral balance, noise, pitch drift, articulation, reverberation-like tails, and brief transitions. These are training perturbations rather than a calibrated simulation of room acoustics. The selected checkpoint was fine-tuned on the repository’s realistic, fast-note, and practice-like synthetic training families; Appendix A.1 records its configuration.

The presence verifier is trained separately on played score notes and planted omissions. Extra- and missed-note classifiers use synthetic development predictions, including score-group-held-out calibration for the inherited gates. Deployment changes to gate thresholds and boundary rules are versioned separately from the acoustic checkpoint. The optional boundary forest uses archived synthetic development controls; those controls were reused during calibration and do not constitute an independent test. The acoustic model, verifier, and alignment procedure are therefore trained or selected in separate stages, without end-to-end gradient optimisation through the current dynamic program. A common, independently annotated evaluation is still required to quantify the effect of these revisions relative to the historical results.

5 Evaluation

This section preserves the previously reported AudioEval experiments and their original model configurations. In particular, the 358-recording synthetic comparison uses the Basic Pitch-based model documented in Appendix A.2. The results, real-recording comparisons, and case studies below predate the current CTC/passage pipeline in Section 4; they are not measurements of that revision. We compare the historical systems with Polytune and LadderSym on synthetic clarinet recordings and on 11 manually labelled recordings of human playing. The synthetic comparison evaluates score correspondence and event labelling; the real-recording comparison evaluates the type and timing of reported errors. Two case studies illustrate how model predictions relate to the written score.

5.1 Experimental Setting

The comparison uses 358 synthetic validation recordings, all drawn from Mozart excerpts. Their reference annotations contain 40,712 events, including correct notes, repeated notes, and errors. This set was available during development, and the reported values are validation results.

The historical AudioEval model uses the 4,544-recording training set described in Section 3.1. We trained Polytune and LadderSym from scratch on a separate corpus of 10,049 synthetic recordings and used 1,116 recordings for model selection. The selected models completed 31 and 26 training epochs, respectively. Exact-audio comparisons identified 21 evaluation recordings in baseline training and three in baseline model selection. The systems share the evaluation set and scoring procedure, with the training and model-selection conditions described above.

All methods analyse complete recordings. Polytune receives practice and reference audio; LadderSym additionally receives a symbolic score prompt. Their timestamped note predictions

are assigned to written locations using the same conversion procedure. This procedure pairs compatible missed and extra predictions into wrong notes, aligns the remaining notes to the score, and checks for repeated passages. Its parameters are fixed for both baselines. Appendix A.3 provides the training and conversion details.

5.2 Evaluation Metrics

The primary measure is *score-based weighted micro-F1*. It assesses whether a predicted event is attached to the correct musical location and given the correct type. A written note is identified by its position in the score; during a repetition, its performed occurrence is also distinguished. An extra note has no written counterpart and is identified by its event in the generated performance.

Predictions and reference events are matched one-to-one. A pair receives credit 1 when both location and type agree, 0.5 when the location agrees but the type differs, and 0 when the location is incorrect or unavailable. Each event can contribute to at most one pair. If C is the total match credit, N_p the number of predicted events, and N_g the number of reference events, then

$$P = \frac{C}{N_p}, \quad R = \frac{C}{N_g}, \quad F_1 = \frac{2C}{N_p + N_g}. \quad (5)$$

The term *weighted* refers to the fractional match credit, and *micro* indicates that credit and event counts are pooled across recordings. Correct, repeated, and erroneous notes all contribute to the measure.

An event type includes its alignment relationship, note label, rhythm status, and repetition status. The baselines supply note-content labels; AudioEval also supplies a rhythm decision. The same complete-event scoring rule is applied to all three systems. Intonation, meaning fine pitch deviation within a note, is outside the comparison. Confidence intervals are estimated with 1,000 bootstrap samples, resampling whole recordings to retain dependence among events in a performance.

5.3 Comparison with Existing Detectors

Table 2 and Figure 2 show the results. AudioEval achieved weighted micro-F1 of 61.04%, compared with 41.39% for Polytune and 34.71% for LadderSym. The absolute gains, computed from unrounded scores, are 19.66 and 26.33 percentage points, respectively.

Table 2: Score-based weighted precision, recall, and micro-F1 on 358 synthetic validation recordings. Values are percentages. Confidence intervals are estimated by resampling recordings 1,000 times.

Method	Precision	Recall	F1	95% F1 interval
Polytune	44.27	38.85	41.39	[39.94, 42.74]
LadderSym	39.52	30.95	34.71	[33.25, 36.07]
AudioEval	55.52	67.79	61.04	[59.37, 62.72]

AudioEval also achieved the highest weighted precision (55.52%) and recall (67.79%).

5.4 Recorded Clarinet Performances

The student author recorded 11 clarinet performances and manually reviewed their error labels. Table 3 reports precision, recall, and micro-F1 for AudioEval, Polytune, and LadderSym. A prediction matches an annotated error when their error types agree and their time intervals overlap sufficiently. We measure this overlap using interval intersection-over-union (IoU): the duration shared by the two intervals divided by their combined duration, counting the overlap

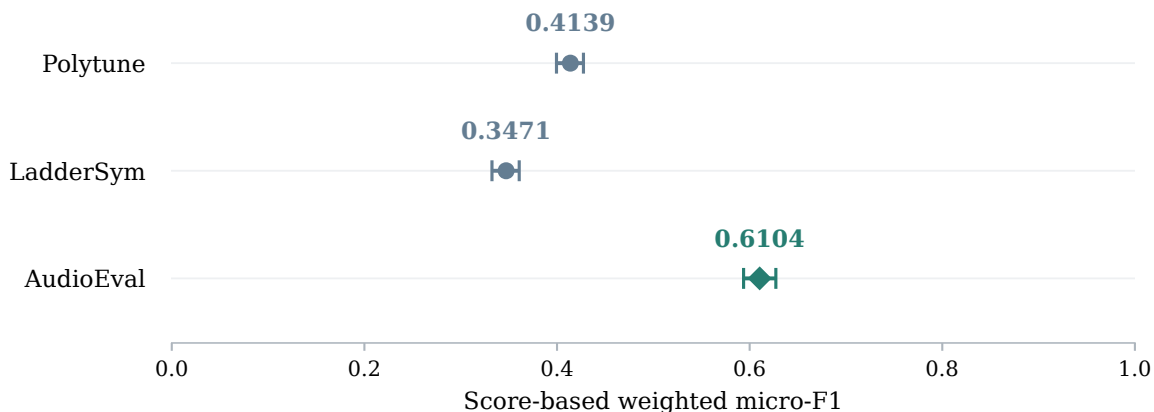


Figure 2: Score-based weighted micro-F1 for the three systems on the same 358 synthetic validation recordings. Markers show point estimates and horizontal lines show 95% recording-bootstrap confidence intervals. Training conditions and the conversion of baseline outputs are described in Section 5.1.

once. A match requires IoU of at least 0.3, and each prediction and annotation can be matched only once.

This error-event measure counts only errors, whereas the synthetic measure also includes correctly played notes and gives partial credit for the correct score location. The two F1 measures therefore answer different questions and should be interpreted separately.

Table 3: Error-event precision, recall, and micro-F1 at interval $\text{IoU} \geq 0.3$ on 11 manually labelled clarinet recordings. Values are percentages. Only error labels are scored.

Method	Precision	Recall	F1
AudioEval	20.26	34.36	25.49
Polytune	1.03	18.31	1.96
LadderSym	1.78	15.49	3.20

The comparison uses the same trained Polytune and LadderSym models as the synthetic experiment. Their outputs are converted to the error categories used here. The annotations contain wrong notes, missed notes, extra notes, rhythm errors, and unintended repetitions; rhythm errors remain in the evaluation even though the baselines do not predict them. Appendix A.6 gives the inference settings and scoring details.

5.5 Case Studies

The following examples show how AudioEval associates repetitions, wrong notes, and extra notes with passages in a recording and score. The first has human annotations for the events discussed; the second illustrates the model’s alignment and reported labels. Pitch names use written clarinet pitches with octave numbers; for example, C4 denotes middle C.

Case 1: unintended repetitions and a wrong note. The first recording covers bars 1–22 of a clarinet arrangement from *Howl’s Moving Castle*. The player completes the opening, pauses, and at 9.8s returns to its first note, E5, to replay bars 1–4. AudioEval identifies this return as an unintended repetition and associates the repeated notes with those same four bars. A human annotator marked the same interval as a repetition. In bar 12, the transcriber reports C6 in

place of the written A5, and AudioEval identifies a wrong note at the location marked by the annotator. The player also returns to the arpeggio in bar 11 and later repeats the figure in bar 15. AudioEval identifies both returns, matching the other two human repetition annotations.

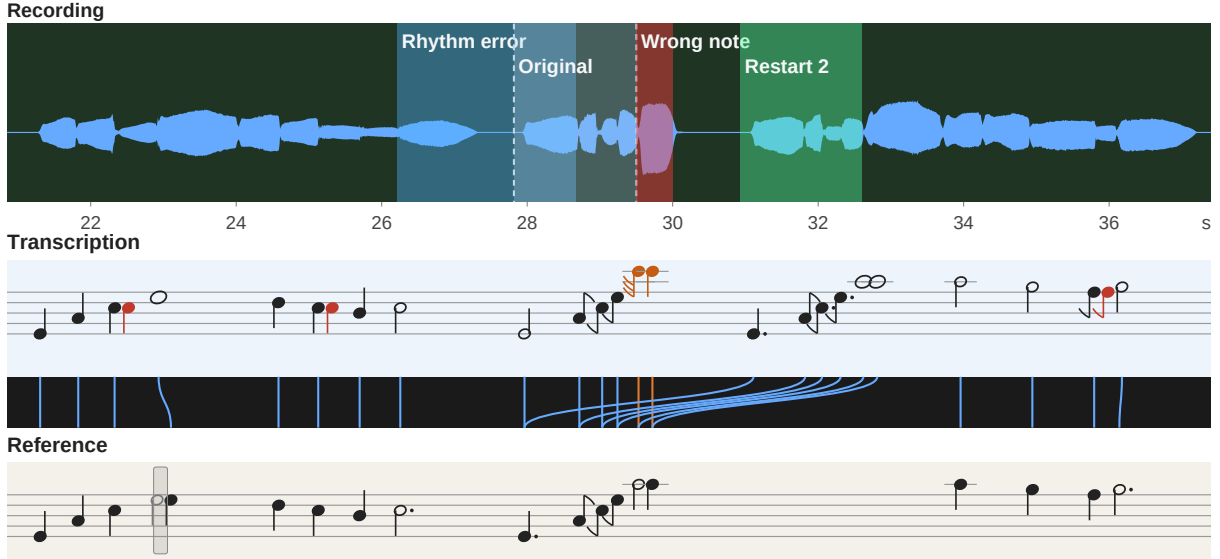


Figure 3: The bar-12 region in Case 1. The waveform shows the recording over time; coloured regions mark model predictions. The upper staff shows transcribed notes and the lower staff shows the reference score. Lines connect corresponding notes. “Original” marks the first attempt, and “Restart 2” marks the second unintended repetition in the recording. Its notes reconnect to the earlier passage in the score.

Tracking each occurrence of a repeated passage allows the feedback to distinguish the first attempt from a later one and replay the relevant interval. The repeated opening remains linked to bars 1–4, while the C6 is identified as a wrong note in bar 12. Figure 3 shows that wrong note and the following repetition. The human annotations corroborate the three repetitions and the wrong note discussed here; the other displayed predictions are not part of that comparison.

Case 2: gaps in the alignment and extra notes. The second recording covers bars 31–36 of a clarinet arrangement of *A Cruel Angel’s Thesis*. The score begins G4–F4–G4–A4. The first note retained by AudioEval is F4 at 2.4s, which is matched to the second written note, leaving the opening G4 unmatched. The alignment also leaves several notes in bars 32–33 unmatched. Two later A4 notes, at 5.3s and 10.0s, have no written counterpart and are reported as extra notes in bars 34 and 35, respectively.

Figure 4 shows how the two extra notes remain between their neighbouring score locations while the surrounding notes continue to follow the score. It also distinguishes alignment from feedback: the grey notes show where no performed note was matched, whereas the coloured waveform regions show the errors actually reported. This example illustrates the model’s output and is not an independent assessment of detection accuracy.

6 MusicPal: Feedback and Continued Data Collection

MusicPal is a proposed desktop practice companion for recording performances, viewing AudioEval’s feedback on the score, and replaying passages that need attention. It is also designed to save recordings for later manual annotation and model development.

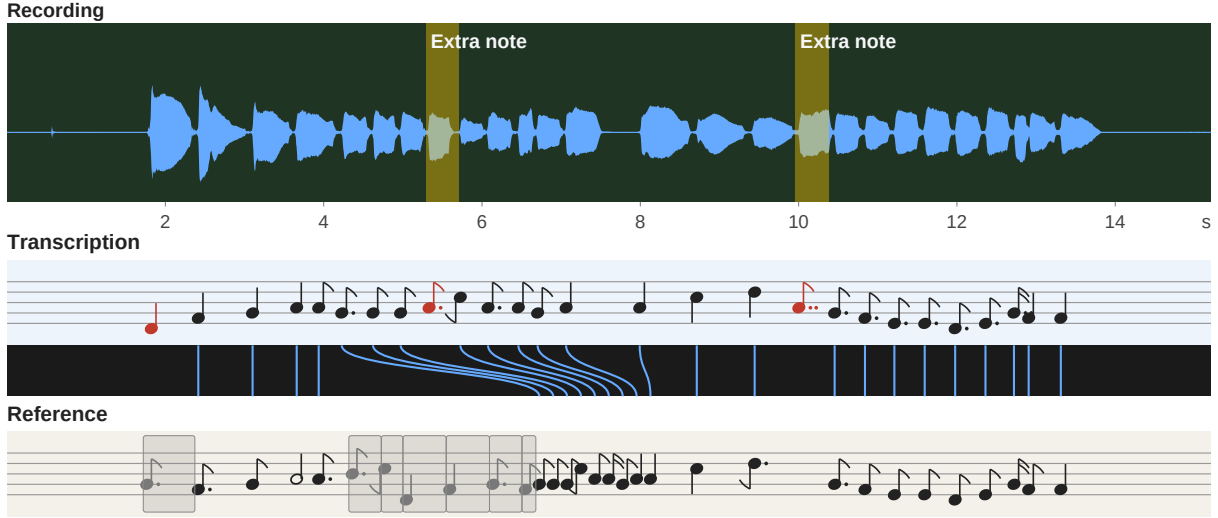


Figure 4: Score alignment and feedback in Case 2. The upper staff shows transcribed notes; extra notes appear in red. The lower staff shows the reference score, with unmatched notes in grey. Lines connect matched notes. Coloured regions on the waveform mark the two reported extra-note errors. Grey notes indicate gaps in the alignment and do not by themselves indicate that a missed-note error was reported. The figure shows model predictions.

6.1 Device and Server Architecture

In the proposed architecture (Figure 5), MusicPal records audio, displays the score, and handles playback, while a remote server runs AudioEval. The server returns error types, written locations, and recording intervals for presentation on the device.

The recording path begins with two microphones and an ES7210 analogue-to-digital converter. Digital audio passes to an ESP32-P4 controller, which buffers the recording and manages the display and storage. The design includes 32 MB of working memory, 32 MB of flash storage, and a microSD card for longer recordings. A companion wireless chip, such as the ESP32-C6, transfers the recording and selected score to the server. Keeping an audio buffer allows capture to proceed independently of screen updates and network transfers.

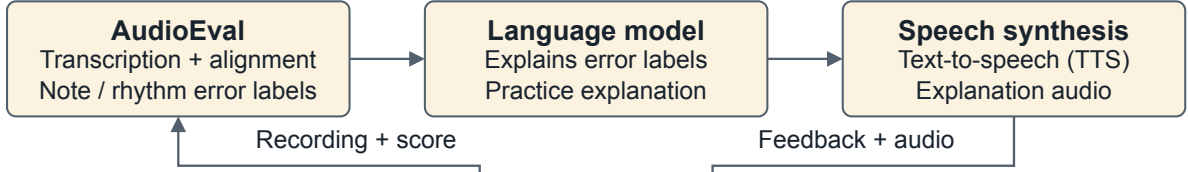
A 4.3-inch touchscreen presents the score and highlights the returned error locations. An ES8311 audio codec and amplifier provide speaker playback, allowing the player to hear the relevant interval. Optional local pitch tracking can display a pitch trace during playing; the musical error labels are produced by AudioEval. The design also includes a small servo for head orientation.

6.2 From Analysis to Practice Feedback

The player would select a score and record a passage. After analysis, MusicPal would highlight the returned error locations and make the corresponding recording intervals available for replay. Missed notes appear at their intended positions, extra notes between neighbouring aligned notes, and unintended repetitions at the repeated passage. The same written location can be associated with more than one recording interval when a passage is played again.

AudioEval would determine which errors to report. An optional language model could use those labels and their associated score locations and recording intervals to produce short explanations, which speech synthesis would read aloud.

(a) Server-side analysis and feedback



(b) MusicPal

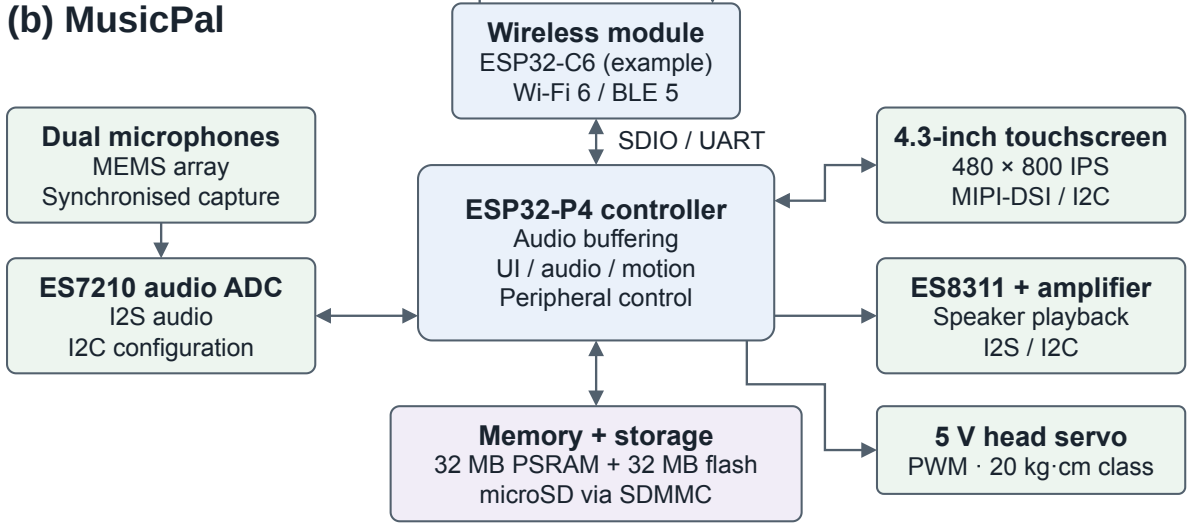


Figure 5: Proposed MusicPal architecture. The device records a performance and sends it with the selected reference score to the analysis server. AudioEval produces error labels linked to the score and recording. MusicPal presents the returned feedback and supports playback. Optional language-model and speech-synthesis components turn the labels into spoken explanations.

6.3 Future Data Collection

Each saved MusicPal session would contain the original audio, its reference score, and the corresponding model predictions. An annotator could then listen to a passage, inspect the proposed labels, and correct their types or locations. Reviewed annotations could contribute to later training collections or to separately maintained evaluation sets.

Over time, these manually reviewed recordings could complement the synthetic training examples and support evaluation on recorded human playing.

7 Conclusion

We present AudioEval, a layered method for clarinet practice feedback. The current method combines dual-resolution CTC transcription, bounded passage and repetition alignment, and acoustic verification of possible errors. It links note errors and repeated passages to canonical score locations and recording intervals, while retaining an explicit uncertain state. Same-pitch boundary restoration remains an experimental extension.

The earlier Basic Pitch-based configuration achieved a score-based weighted micro-F1 of 61.04% on 358 synthetic validation recordings, exceeding Polytune and LadderSym by 19.66 and 26.33 percentage points, respectively. It also obtained the highest precision and recall among the compared systems. On 11 manually labelled clarinet recordings, AudioEval achieved an error-event micro-F1 of 25.49%, compared with 1.96% for Polytune and 3.20% for LadderSym. These earlier results and case studies do not measure the revised pipeline; a matched evaluation is needed to establish its comparative performance. The case studies illustrate how repetitions

and note errors can be associated with specific score passages and recording intervals. The proposed MusicPal device would present this feedback alongside the score and recorded sound, while saving practice recordings for future manual annotation and model development.

Limitations

The revised methodology and the existing result tables refer to different configurations. The dual-resolution CTC transcriber, bounded passage runtime, and optional v10 refinement require a new matched evaluation before competitive performance can be claimed from those tables. The current exporter covers note-content errors and repetitions, without the earlier rhythm classifier. Same-pitch merging can erase real attacks; v10 can restore only already decoded, structurally supported boundaries and has been calibrated on reused development controls. Passage retrieval can choose the wrong occurrence of an identical phrase, and its earliest-plausible-location policy does not resolve that uncertainty acoustically. Bounded windows exclude long unsegmented performances and unrestricted jumps. Error gates trade recall for fewer false alarms, and their scores are not calibrated probabilities.

The synthetic comparison uses validation recordings under the training conditions reported in Section 5.1. The systems differ in training data and the labels they provide. The real-recording comparison covers 11 performances by the student author and uses an error-event measure, so its F1 values should be interpreted separately from the synthetic score-based results. The case studies illustrate particular model decisions: the first includes human annotations for the events discussed, while the second shows model-generated labels. Evaluation across more players, pieces, and recording conditions, with independent annotations and separate development and evaluation data, would establish how broadly the results apply. Further experiments could vary transcription, score alignment, and error labelling one component at a time on the same data.

MusicPal remains at the design stage. Implementation and integration will be followed by tests conducted by the student author across microphone distances, room acoustics, recording levels, and network conditions. Simultaneous recordings with a reference device will support comparisons of capture quality and prediction consistency. These tests will also measure response time and whether complete recording and feedback sessions run reliably over repeated trials.

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A Implementation and Evaluation Details

A.1 Current Transcriber and Aligner Configuration

The following settings describe the current method, not the historical result tables. The selected acoustic checkpoint is `precision-v4/v5-dclike/best.pt`; v9 and v10 reuse this checkpoint rather than the later pilot transcriber. It uses 32 spectral channels, temporal width 256, twelve dilated blocks, and dropout 0.10. The long mel branch covers 45–8,000 Hz and the short branch 100–8,000 Hz. Inference uses 4,096-frame windows with 1,024-frame overlap.

The final fine-tuning stage starts from `precision-v4/v4-realism/best.pt` and combines the `realistic92`, `fast102`, and `dclike11` training caches. It runs three epochs with batch size eight, 2,048-frame crops, AdamW weight decay 10^{-3} , and a one-cycle schedule peaking at 1.5×10^{-4} . Gradient norm is clipped at 2.0; the seed is 20260927. The selected checkpoint is from epoch two, using the recorded validation selection score. These are the final fine-tuning settings, not the full pretraining history.

The default runtime is `v9-passage-v2`, using the v9 candidate’s decoder and same-pitch settings and the bounded v5 aligner. It inherits the v6 acoustic verifier and logistic gate coefficients, with the missed-note threshold changed to 0.80; the extra-note threshold remains 0.85. Table 4 records the main operating parameters. Hashes in the candidate manifests identify the checkpoint and versioned components.

Table 4: Current decoding and alignment settings. These are operating parameters, not measured accuracies.

Component	Setting
CTC candidates	Blank scale 0.5; optional peak threshold 0.08; alternative-pitch support at least 0.10.
Transition candidates	At most 45 ms to the next primary onset and attack support below 0.35.
Passage retrieval	Reliable-note confidence at least 0.20; at least eight reliable notes for long-score retrieval; similarity at least 0.52. Scores with at most 128 notes bypass retrieval.
Passage ranking	Alignment cost per candidate plus $0.4(1 - \text{similarity})$. Earliest plausible location within similarity difference 0.04 and rank difference 0.08 of the best.
Search bounds	Four windows; 512 score notes per window; 1,024 repeat hypotheses per window; 2,048 expanded units and two million DP cells per hypothesis; 20 million locator cells.
Base edit costs	Substitution 1.1; insertion scale 1.2; deletion 1.0; merge 0.15; repeat-copy penalty 1.0; alternative-pitch penalty 0.6.
Timing refinement	Anchor confidence 0.90; $\lambda_t = 0.2$, $\kappa = 1$, and $\tau_j = \max(0.06 \text{ s}, 0.35 \hat{d}_j)$ for expected duration $\hat{d}_j$.
Error verification	Extra threshold 0.85; missed threshold 0.80; whole-clip match/copy fraction at least 0.45. Boundary-context notes outside audible coverage are not automatically labelled missed.

The v9 same-pitch rule examines up to 90 ms on each side, clipped at neighbouring onsets with a 5 ms margin. Its gap veto uses an 18 dB depth and a configured 10 ms interruption target. The observed low-energy core is shorter because the RMS window spans 4 ms; the rule compensates for this window and sampling phase rather than treating the core as an exact silence-duration estimate. The permissive energy-range limit is 54 dB, with normalised mel step and drift bounds of 0.35 and 0.65; model attack and voicing vetoes are disabled in this candidate. These settings were calibrated using injected split controls and are known to merge genuine repeated notes.

The separate `v10-boundary-v1` configuration adds a 96-tree boundary forest with maximum depth five, minimum leaf size three, and seed 20261005. It requires original primary-candidate confidence at least 0.5 and a repeat vote at least 0.5, followed by the realignment checks in Section 4.3. Its 442 usable archived acoustic rows come from synthetic development controls. The control material was reused during calibration, including exploratory inspection of the check subset; neither the control replay nor unchanged sparse error labels establishes independent boundary accuracy.

A.2 Historical AudioEval Configuration

The AudioEval model used in the synthetic comparison uses Basic Pitch as a fixed transcriber and a learned scoring function to rank its note candidates. The alignment model represents acoustic, score, and repetition features separately before combining them for path scoring and error classification. The input to each alignment decision contains 26 features. The path scorer uses 32-dimensional component representations and a hidden dimension of 64; the additional acoustic, score, and repetition representations have dimension 64 and use hidden dimension 128. Dropout is 0.08.

The alignment search retains at most 12 candidate score spans per note candidate and 48 states at each step. A span covers at most four consecutive written events, and an ordinary transition can skip at most 24 score events. Continuation evidence examines up to three subsequent candidates, allowing one skipped candidate and one skipped score event. These bounds limit the number of alternatives considered when decoding a recording.

We estimate the tempo scale by dividing the time span of the acoustic candidates by the duration of the reference score in quarter-note units, restricting the result to 0.08–2.5 seconds per quarter note. Duration and note-spacing features are expressed relative to this scale. The rhythm classifier uses a probability threshold of 0.5 in the reported evaluation. The note classifier chooses among correct, wrong, and extra for retained notes; missed notes come from the skipped score events.

The system is initialised with a trained alignment scorer. This initial scorer remains fixed while additional acoustic, structural, and error-classification components are trained. Training comprises two epochs for acoustic decisions, five for repetition structure, five for error classification, and ten for joint optimisation of the trainable components. AdamW uses a learning rate of 2×10^{-4} and weight decay 10^{-4} . The pretrained Basic Pitch weights remain fixed throughout. Synthetic training and validation contain 4,544 and 358 recordings, respectively; the reserved partition of 4,022 recordings is outside the reported comparison.

A.3 Baseline Training

Polytune and LadderSym are trained from scratch on 10,049 synthetic recordings, with 1,116 recordings for model selection. The models used in the comparison completed 31 and 26 epochs, respectively. Both use seed 365, peak learning rate 2×10^{-5} , 4,000 warmup updates, a cosine schedule configured for 40 epochs, and gradient accumulation over 16 training steps. Training samples 2.048-second audio segments; evaluation covers complete recordings.

Inference uses 16 kHz mono audio, 32-bit floating-point computation, greedy decoding, and a maximum of 1,024 output tokens. LadderSym additionally receives a reference-score prompt of up to 1,024 tokens. The same 358 evaluation recordings and 40,712 reference events are used for all three systems. Exact-audio overlap with baseline training and model selection is reported in Section 5.1.

A.4 Assigning Baseline Predictions to Score Locations

The baselines produce timestamped notes with the native labels **Correct**, **Missing**, and **Extra**. We use *missed note* in the paper for the category denoted by **Missing**. The same procedure assigns both baselines’ predictions to score locations before evaluation.

An extra-note prediction and a missed-note prediction form a wrong-note prediction when their pitches differ, they are mutually nearest in onset time, and their onsets are within 100 ms. Correct notes, these paired predictions, and remaining missed notes are then aligned monotonically to the reference score. The edit costs are 1 for insertion or deletion and 1.5 for a pitch mismatch. A paired wrong-note prediction uses the expected pitch of the missed note to determine its score location. Gaps introduced by conversion do not create additional missed-note predictions.

A remaining sequence of at least three extra notes may be assigned to a repetition if its pitches agree with at least 80% of a previously visited contiguous score sequence. The longest qualifying sequence is selected; ties favour pitch agreement and then the most recent score position. Previous predicted visits determine the performed occurrence assigned to the repeated notes.

The conversion receives only model predictions and the reference score. Its parameters were fixed before scoring. Unlocated and duplicate predictions remain in the prediction count, and no performance-event identity is assigned without evidence from the prediction. The baselines provide no rhythm-error decision, so their rhythm status is set to absent under the complete-event scoring rule. Intonation is excluded for every system.

A.5 Scoring and Confidence Intervals

Each system’s predictions are saved before evaluation annotations are read. The one-to-one matcher selects the assignment with maximum total credit under the location and type rules in Section 5. Credit and event counts are pooled across recordings before precision, recall, and F1 are calculated.

Confidence intervals use 1,000 bootstrap replicates. Each replicate samples 358 recordings with replacement and recomputes the pooled weighted micro-F1. The interval endpoints are the 2.5th and 97.5th percentiles of the resulting values. Resampling recordings preserves the grouping of notes within a performance. Differences between systems are calculated from the unrounded point estimates.

A.6 Real-Recording Evaluation

The real-recording comparison uses the previously selected Polytune and LadderSym models after 31 and 26 training epochs, respectively. Both analyse complete recordings with 16 kHz mono inputs and bfloat16, a 16-bit floating-point format. Recordings are processed one at a time, with a maximum of 1,024 output tokens. LadderSym also receives a fixed symbolic encoding of the reference score. Predicted error intervals are evaluated individually, without merging adjacent intervals.

The annotations contain 71 error events across the 11 recordings. Matching requires the same error type and interval IoU of at least 0.3, with one-to-one assignment. Each match receives one credit; no partial credit or separate pitch-matching condition is applied. Precision, recall, and micro-F1 are computed from counts pooled across recordings. Polytune matches 13 reference events with 1,258 predictions; LadderSym matches 11 with 617 predictions. Rhythm errors remain in the reference annotations even though the baselines do not predict that category.